Case Study 28: Repeated Transaction Detection

1. Problem Statement

Process a list of transaction records containing transaction IDs, user accounts, and amounts, detect duplicate transaction IDs or identical repeated transfers within short logs, flag high-value threshold violations, and generate a list of suspicious entries.

2. Objectives

- Traverse transaction lists using loops and conditional logic to identify duplicate transaction identifiers.
- Detect identical repeated transfers (same account, same amount) within a batch.
- Flag transactions exceeding safety amount thresholds as suspicious for audit inspection.

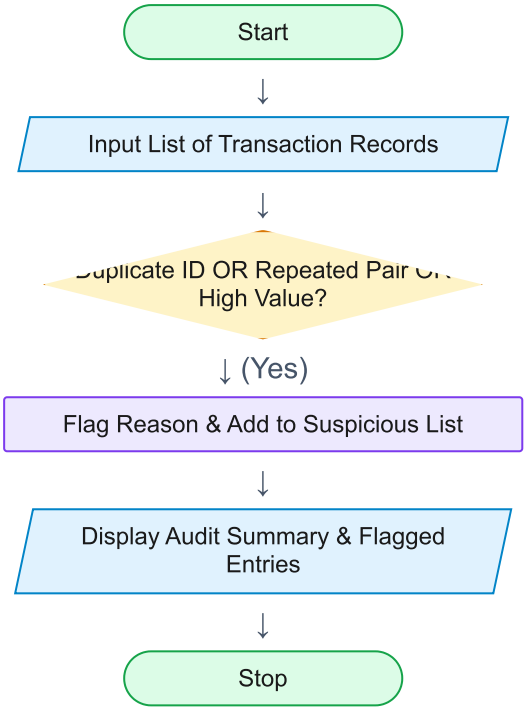
3. Assumptions

- High-value transaction threshold is set to **Rs. 50,000.00**.
- Duplicate IDs are illegal entries in a transaction stream.
- Repeated identical transactions from the same account with identical amounts within the log are flagged as potential duplicate processing errors.

4. Algorithm

1. Start.
2. Read transaction dataset `[ (txn_id, account, amount), ... ]`.
3. Initialize empty sets/lists: `seen_ids`, `seen_transfers`, and `suspicious_list`.
4. For each transaction record `(txn_id, account, amount)`:
    - If `txn_id` exists in `seen_ids`, flag as **DUPLICATE ID**.
    - Else add `txn_id` to `seen_ids`.
    - If `(account, amount)` pair exists in `seen_transfers`, flag as **REPEATED TRANSFER**.
    - Else add `(account, amount)` to `seen_transfers`.
    - If `amount >= 50000.0`, flag as **HIGH VALUE EXCEEDED**.
5. If any flag condition is met, append record and reasons to `suspicious_list`.
6. Display complete transaction status ledger and detailed suspicious audit report.
7. Stop.

5. Flowchart



6. Pseudocode

```
START INPUT transactions threshold = 50000.0 seen_ids = [] seen_pairs = []
suspicious = [] FOR EACH txn IN transactions DO txn_id = txn[0] acc = txn[1] amt =
txn[2] reasons = [] IF txn_id IN seen_ids THEN APPEND "Duplicate Txn ID" TO reasons
ELSE APPEND txn_id TO seen_ids ENDIF pair = (acc, amt) IF pair IN seen_pairs THEN
APPEND "Repeated Account & Amount Transfer" TO reasons ELSE APPEND pair TO
seen_pairs ENDIF IF amt >= threshold THEN APPEND "High Value Threshold Violation (≥
50k)" TO reasons ENDIF IF length(reasons) > 0 THEN APPEND (txn_id, acc, amt,
reasons) TO suspicious ENDIF ENDFOR PRINT suspicious STOP
```

7. Complete Python Source Code

```
     # Repeated Transaction Detection def audit_transactions(transactions_list, threshold=50000.0): seen_ids = [] seen_pairs = [] suspicious_entries = [] print("=" * 72) print(" TRANSACTION AUDIT & ANOMALY DETECTION REPORT") print("=" * 72) print(f"{'Txn ID':<10}{'Account':<14}{'Amount (Rs.)':<16}{'Status':<14}{'Audit Findings':<18}") print("-" * 72) for txn in transactions_list: t_id, acc, amt = txn[0], txn[1], txn[2] reasons = [] # Check 1: Duplicate Transaction ID if t_id in seen_ids: reasons.append("Duplicate Txn ID") else: seen_ids.append(t_id) # Check 2: Repeated Account & Amount Pair pair = (acc, amt) if pair in seen_pairs: reasons.append("Repeated Transfer Pair") else: seen_pairs.append(pair) # Check 3: High Value Threshold if amt >= threshold: reasons.append("High-Value Limit Exceeded") status = "SUSPICIOUS" if reasons else "NORMAL" findings = ", ".join(reasons) if reasons else "Clean" if reasons: suspicious_entries.append((t_id, acc, amt, reasons)) print(f"{t_id:<10}{acc:<14}{amt:<16,.2f}{status:<14}{findings:<18}") print("-" * 72) print(f"Total Transactions Processed : {len(transactions_list)}") print(f"Suspicious Entries Flagged : {len(suspicious_entries)}") print("=" * 72) # Sample Batch Dataset txns = [ ("TXN1001", "ACC_401", 15000.0), ("TXN1002", "ACC_402", 62000.0), # High value ("TXN1003", "ACC_403", 8500.0), ("TXN1001", "ACC_404", 1200.0), # Duplicate ID ("TXN1005", "ACC_401", 15000.0), # Repeated pair (ACC_401, 15000) ("TXN1006", "ACC_405", 55000.0) # High value ] audit_transactions(txns)
```

8. Test Case and Expected Output

Execution Results for Audit Batch:

```
     ========================================================================
TRANSACTION AUDIT & ANOMALY DETECTION REPORT
======================================================================== Txn ID
Account Amount (Rs.) Status Audit Findings ----------------------------------------
------------------------------ TXN1001 ACC_401 15,000.00 NORMAL Clean TXN1002
ACC_402 62,000.00 SUSPICIOUS High-Value Limit Exceeded TXN1003 ACC_403 8,500.00
NORMAL Clean TXN1001 ACC_404 1,200.00 SUSPICIOUS Duplicate Txn ID TXN1005 ACC_401
15,000.00 SUSPICIOUS Repeated Transfer Pair TXN1006 ACC_405 55,000.00 SUSPICIOUS
High-Value Limit Exceeded ----------------------------------------------------------
-------------- Total Transactions Processed : 6 Suspicious Entries Flagged : 4
========================================================================
```

9. Conclusion

The Repeated Transaction Detection engine successfully parses transaction streams, identifies duplicate transaction identifiers and identical transfer pairs, and flags high-value transfers to protect financial systems from duplicate processing and compliance violations.